

Question: 1

Install and configure a basic Apache or Nginx web server on your local machine, then verify it is running by accessing <http://localhost> in your browser?

Answer: 1. Installation Phase

For Linux (Ubuntu/Debian):

Open your terminal and run:

```
bash
```

```
sudo apt update && sudo apt install nginx -y
```

Use code with caution.

For Windows:

Download the Nginx Windows zip archive from the [official download page](#).

Extract the folder contents to your preferred directory (e.g., C:\nginx).

Open Command Prompt as an administrator, navigate to the folder, and run:

```
cmd
```

```
cd C:\nginx
```

```
start nginx
```

Use code with caution.

2. Service Management (Start & Status Control)

For Linux:

Start Service: `sudo systemctl start nginx`

Verify Active Status: `sudo systemctl status nginx`

For Windows:

Check if the running tasks are active in the background using the tasklist utility:

cmd

```
tasklist /fi "imagename eq nginx.exe"
```

Use code with caution.

3. Verification Phase

Open any web browser (Chrome, Edge, or Firefox).

Enter this exact string in the URL path window: <http://localhost> (or <http://127.0.0.1>) and press Enter.

Expected Result: The page will display the default server welcome splash banner, reading exactly: "Welcome to nginx!".

Question: 2

Use Wireshark to capture HTTP traffic while loading your web server homepage, then identify and list the source and destination IP addresses and the HTTP request method used?

Answer: 1. How to Isolate the Web Traffic

Filter: Type http in the top display filter bar and press Enter.

Selection: Look for the row that has GET / HTTP/1.1 in the Info column and click it.

2. Captured Packet Parameters

Expand the Internet Protocol (IPv4) and Hypertext Transfer Protocol (HTTP) sections in the middle details pane to document these exact values:

Source IP Address: 127.0.0.1 (or your local client machine's LAN IP, such as 192.168.1.100).

Destination IP Address: 127.0.0.1 (The loopback localhost IP where your Apache/Nginx web server is hosting the pages).

HTTP Request Method: GET (The standard method used by browsers to retrieve web page files like index.html from a server).

Question: 3

Transfer a small text file from your computer to a remote system using FileZilla (FTP client) with both FTP and SFTP, and capture the network traffic for each protocol in Wireshark. Compare and describe the difference in packet details between FTP and SFTP.
Hint: Look for visible file content or credentials in FTP packets versus encrypted data in SFTP packets?

Answer: 1. Wireshark Capture Parameters

FTP Filter: Type ftp or ftp-data in the top display filter bar.

SFTP Filter: Type ssh in the top display filter bar (since SFTP runs inside an encrypted SSH tunnel over port 22).

2. Side-by-Side Packet Details Comparison

Analysis Metric	FTP Capture (port 21)	SFTP Capture (port 22)
Authentication / Login Security	Plain Text Exposure: The login username (USER) and password (PASS) are completely visible in plain text inside the packet payload.	Fully Encrypted: No login metrics or user passwords can be read. The entire handshake happens inside an encrypted secure shell framework.

Command Execution Visibility	Visible Commands: All standard directory traversal interactions and commands (like SYST, PWD, PORT, STOR) show up as raw readable text.	Obfuscated Stream: All internal file system interactions are packaged into secure generic SSH packets labeled as Encrypted Request Packet.
File Content Transmission	Raw Data Stream: Opening the TCP Stream reveals the exact text contents of the transferred file in plain text.	Cipher Block Data: Opening the stream shows unreadable cryptographic garbage data generated by the active cipher block.
Data Integrity Verification	None: Packets can be intercepted, read, or modified mid-transit via standard network spoofing attacks without triggering internal alerts.	Cryptographic Hash Control: Built-in cryptographic MAC (Message Authentication Code) protocols prevent active packet manipulation.

3. Key Laboratory Takeaway

FTP transfers all structural payloads in clear text, making it highly vulnerable to packet sniffing tools like Wireshark.

SFTP wraps the entire session inside an encrypted SSH container, ensuring absolute confidentiality for files, tracking structures, and user passwords.

Question: 4

Analyze a public website (such as <https://www.flipkart.com>) in Wireshark and find a packet that uses HTTPS. Take a screenshot of the packet details and explain why you cannot view the actual data being sent?

Answer: 1. How to Find the HTTPS Packet in Wireshark

Filter: Type `tls` or `tcp.port == 443` in the top display filter bar and press Enter (HTTPS runs over TLS/SSL encryption on port 443).

Selection: Look for rows in the packet list labeled Application Data or Encrypted Handshake Message in the Info column and click on one.

2. Expanded Protocol Details (For Screenshot & Annotation)

When you capture traffic from a public website like Flipkart, expand the protocol window in the middle details pane to see the following structural blocks:

Frame / Ethernet II: Data Link Layer (MAC addresses).

Internet Protocol Version 4 (IPv4): Network Layer (Source/Destination IP addresses).

Transmission Control Protocol (TCP): Transport Layer (Destination Port: 443).

Transport Layer Security (TLS): Application Security Layer (Shows fields like TLSv1.3 Record Layer: Encrypted Application Data).

3. Why You Cannot View the Actual Data Being Sent

Transport Layer Security (TLS) Encryption: Flipkart uses HTTPS, which encrypts the entire application layer payload using modern cryptographic protocols (TLS 1.2 or TLS 1.3).

Asymmetric & Symmetric Cryptography: During the initial connection handshake, your browser and Flipkart's servers securely exchange unique symmetric session keys. All subsequent data (like your login credentials, searches, or shopping cart items) is scrambled using these keys before leaving your device.

Ciphertext Payload: In Wireshark's lower packet bytes pane, the raw payload appears strictly as randomized, unreadable hexadecimal garbage values (ciphertext). Without possessing the exact private session key generated in memory during that specific browser session, the data cannot be decrypted by an outside observer.

Question: 5

Write a short protocol analysis report (200-300 words) summarizing what you observed about HTTP, HTTPS, FTP, and SFTP traffic in Wireshark, focusing on how encryption changes what you can see in the packet captures?

Answer: Protocol Analysis Report: The Impact of Encryption on Packet Visibility

This analysis evaluates network traffic captures in Wireshark for HTTP, HTTPS, FTP, and SFTP, focusing on how cryptographic protocols alter data visibility within the packet payload.

Unencrypted Traffic (HTTP and FTP)

When analyzing standard HTTP (Port 80) and FTP (Port 21) streams, Wireshark captures display complete structural transparency. Because these legacy protocols transmit data in clear plain text, an outside observer can fully reconstruct the session payload.

FTP Analysis: The packet bytes pane exposes sensitive metadata explicitly, including plain text credentials (USER and PASS commands), directory pathways, and raw transferred file contents.

HTTP Analysis: The Application Layer displays exposed system headers, request methods (like GET or POST), host domains, and unencrypted web form inputs.

Encrypted Traffic (HTTPS and SFTP)

Conversely, introducing a security layer completely alters packet visibility by obfuscating the application payload while leaving lower-level routing wrappers intact.

HTTPS Analysis: Running over TLS (Port 443), the initial cryptographic handshake remains visible. However, once a secure tunnel is established, all subsequent application traffic is designated as Encrypted Application Data. The underlying content appears strictly as randomized, unreadable hexadecimal ciphertext.

SFTP Analysis: Operating inside an SSH container (Port 22), SFTP completely scrambles authentication sequences, command metrics, and file data.

Structural Summary

While encryption seals the Application Layer payload against unauthorized inspection, it does not hide lower-level headers. In both HTTPS and SFTP captures, Network and Transport headers remain fully visible. Wireshark can still read Source/Destination IP addresses (Layer 3) and Port numbers (Layer 4) to route packets across the infrastructure, but the actual data inside remains completely secure.